FINAL PROBLEM STATEMENTS

1. AI / ML

1.UrbanIncident Fusion & Causal Intelligence

Brief Explanation / Real-WorldContext

Cities continuously generate fragmented signals: trafficcameras, citizen complaints, weather feeds, emergency calls, water-level sensors, road sensorsand public-transport telemetry. The hard problem is not detecting each event independently; itisdeciding which observations belong to the same evolving incident and what is most likely causingit.

For example, a sudden rise in water level, slowing traffic,repeated citizen reports and a blocked-road sensor may all be symptoms of one drainage failure.Aconventional dashboard shows four alerts. An operational system must infer one incident,preserve uncertainty, and update its understanding as new evidence arrives.

Core Challenge

Design a multimodal intelligence engine that ingests heterogeneous time-stamped events and constructs an evolving incident graph. The system should perform event correlation, anomaly detection, severity estimation, probable-cause inference and confidence-aware summarization.

A strong prototype should demonstrate how the same incident is reconstructed when signals arrive out of order, contain duplicates, disagree, or disappear temporarily.

Expected 24-Hour Prototype

Expected prototype: event ingestion layer; temporal/geospatial correlation; incident clustering; causal/evidence graph; severity model; confidence score; operator-facing explanation; API for downstream systems.

Scope, Constraints & Convergence

24-hour teams are not expected to solve general causal inference. They should define a constrained incident vocabulary and prove the architecture on a synthetic or curated real-world dataset.

Convergence: IoT supplies trusted sensor events; Cybersecurity verifies source/device integrity; Web provides the incident command interface.

2.Multimodal Synthetic-Media Forensics & Provenance Reasoning

Brief Explanation / Real-World Context

Synthetic audio, video and imagery are increasingly being used inmisinformation, fraud, impersonation and digital investigations. A simple real/fake classification is insufficient for forensic analysis, because investigators need to understand *what evidencesuggests manipulation, where inconsistencies occur, how the media was processed, and whetherits claimed origin can be trusted*.

A media file may also be partially manipulated for example, genuinefootage may contain a synthetically generated voice, a replaced region, altered timestamps, or multiple generations of compression and editing.

Core Challenge

Develop a multimodal forensic investigation system that analyses media as a collection of interconnected evidence rather than treating it as a binary classification problem.

The system should correlate:

- Visual inconsistencies and manipulation artifacts
- Audio characteristics and synthetic-speech indicators
- Audio–video synchronization
- Frame-level and temporal anomalies
- Metadata and encoding history
- Compression and re-encoding patterns
- File modification history
- Source/device information where available
- Cryptographic hashes and provenance records
- Relationships between multiple copies or versions of the same media

The system should generate an evidence-backed forensic assessment, showing *why* particular portions of the media are suspicious and how different pieces of evidence support or contradict each other.

Expected 24-Hour Prototype

Teams should demonstrate a working vertical prototype containing:

Media Ingestion → Forensic Analysis → Evidence Correlation → Timeline/Region Analysis → Provenance Reconstruction → Investigator Review

The prototype should include at least:

- Multimodal media ingestion
- Multiple independent forensic checks
- Frame/segment-level anomaly identification
- Audio/video consistency analysis
- Metadata and encoding inspection
- Evidence correlation and confidence estimation
- Visual forensic timeline or suspicious-region representation
- Provenance/hash verification
- Human investigator review and annotation

Important Requirement

The solution must not depend primarily on an external deepfake-detection API or third-party "fake/real" classifier.

If external pretrained models or forensic libraries are used, they should function as individual evidence sources within the larger reasoning pipeline, not as the final decision-maker.

Convergence: Cybersecurity/Blockchain can anchor hashes and provenance; IoT can provide trusted capture-device identity; Web provides an investigator workspace.

3. Cross-Source Operations Copilot with Evidence-Backed Reasoning

Brief Explanation / Real-World Context
Operations teams increasingly have information scattered across tickets, logs, sensor streams, documents and alerts. Generative AI can summarize information,but summaries without provenance can hide contradictions and invent connections.

Core Challenge
Build an AI operations copilot that answers operational questions by retrieving and correlating structured and unstructured evidence, identifying conflicts and producing recommendations with source references and confidence.

Expected 24-Hour Prototype
Expected prototype: retrieval layer; structured-event correlation; evidence graph; grounded generation; contradiction detection; recommendation module; audit trail.

Scope, Constraints & Convergence
The challenge is not to build a generic chatbot. The evaluation should focus on grounding, traceability and correct uncertainty handling.

Convergence: all three other domains can provide data or consume recommendations.

2. Cybersecurity & Blockchain

1.Privacy-Preserving Verifiable Identity Network

Brief Explanation / Real-World Context
Organizations repeatedly ask users for full identity documentseven when they only need one attribute. This increases breach impact and creates unnecessary data copies. Build a system in which a user can prove a claim without disclosing unrelated information.

Core Challenge
Design a verifiable-credential architecture supporting issuance, presentation, selective disclosure, revocation and verifier authentication. Demonstrate a realistic claim such as eligibility, age band, employment status or institutional membership.

Expected 24-Hour Prototype
Expected prototype: issuer/holder/verifier roles; cryptographic proof; revocation/status mechanism; wallet-like interface; verifier API; audit events.

Scope, Constraints & Convergence
Do not place raw personal information on a public ledger. Teams must document trust assumptions and key recovery.
Convergence: Web provides identity workflows; AI can detect suspicious credential patterns; IoT can bind credentials to authorized devices/assets.

2.Physical-to-Digital Anti-Counterfeit Trust Fabric

Brief Explanation / Real-World Context
A QR code alone does not prove that a physical product is genuine. A counterfeit item can copy the

identifier of a genuine item. The real challenge is connecting physical observations, ownership transfers and trusted records into a system that can detect impossible histories.

Core Challenge
Build a provenance network where products/lots receive identities and every authorized transfer or scan becomes a verifiable event. Detect duplicate identities, impossible travel and suspicious handovers.

Expected 24-Hour Prototype
Expected prototype: product identity; event signing; ledger or tamper-evident store; verifier; transfer workflow; anomaly API; provenance timeline.

Scope, Constraints & Convergence
Blockchain should solve a trust/audit problem, not act as a fashionable database. Explain what belongs on-chain and off-chain.

Convergence: IoT supplies scans/location/condition data; AI detects movement anomalies; Web provides the supply-chain control tower.

3.Software Dependency Trust Graph & Compromise Detection

Brief Explanation / Real-World Context
Modern applications depend on deep chains of third-party packages. A compromise may occur far below the application's direct dependencies. Teams need visibility into dependency relationships and suspicious changes rather than a static list of known vulnerabilities.

Core Challenge
Build a dependency intelligence system that constructs a graph, tracks version changes, identifies unusual publisher/release behaviour and estimates supply-chain risk.

Expected 24-Hour Prototype
Expected prototype: dependency ingestion; graph construction; version/change comparison; risk scoring; suspicious-change detection; developer-facing report.

Scope, Constraints & Convergence
Use public package metadata or curated datasets. Do not execute untrusted packages.

Convergence: AI can rank suspicious dependencies; Web provides the developer console; blockchain can anchor trusted release metadata.

3. Web & Product Development

1.National-Scale Incident Command Operating Layer

Brief Explanation / Real-World Context
Emergency information is commonly fragmented between phone calls, messaging apps, spreadsheets, dashboards and departmental systems. The challenge is not building another dashboard; it is creating a workflow that preserves provenance, uncertainty and responsibility while the situation changes rapidly.

Core Challenge
Build a role-based incident command platform that accepts heterogeneous events, turns them into cases, assigns responsibility, maintains timelines, supports evidence and exposes a live operational picture.

Expected 24-Hour Prototype
Expected prototype: incident lifecycle; role/permission system; event ingestion API; map/timeline; task assignment; escalation; evidence references; audit trail; real-time updates.

Scope, Constraints & Convergence
The interface should remain usable when dozens of events arrive together. Avoid hard-coding the demo around one incident.

Convergence: AI prioritizes events; IoT supplies live telemetry; Cybersecurityprovides identity, trust and audit

2.Supply-Chain Control Tower with Provenance Reasoning

Brief Explanation / Real-World Context
Supply-chain dashboards often show where goods are, but not whether thehistory makes sense. A control tower should expose missing events, impossible movement, condition breaches and suspicious handovers while keeping the raw event history available for audit.

Core Challenge
Build a web control tower for a constrained supply chain that reconstructs product/lot journeys and highlights anomalies with evidence.

Expected 24-Hour Prototype
Expected prototype: product/lot model; event ingestion; timeline; map; anomaly flags; provenance verification; role-based views; investigation workflow.

Scope, Constraints & Convergence
Do not make the blockchain layer decorative. Teams must define the exact trust property it adds.

Convergence: IoT provides scans and condition telemetry; AI identifies anomalies; Cybersecurity validates actors and events.

3.Digital Handoff & Continuity Management Platform

Brief Explanation / Real-World Context
When a task, case, project, or customer is transferred between employees or teams, important context is often scattered across emails, documents, chats, tickets, and personal notes. This can result in repeated work, missed deadlines, incorrect decisions, and delays in resolution.

Core Challenge
Develop a product that ensures **continuity of information and responsibility during handoffs**. The system should create a structured context package containing relevant information, previous actions, pending tasks, important decisions, supporting documents, and deadlines. The receiving person/team should be able to quickly understand the current state and continue the work without depending on the previous owner.

The product should also identify **incomplete information or unresolved actions before a handoff is completed** and maintain accountability throughout the transition.

Expected 24-Hour Prototype
The prototype should demonstrate:

- Case/project creation and ownership
- Centralized context and document management
- Pending-action and deadline tracking
- Structured handoff workflow
- Handoff completeness/verification
- New-owner acknowledgement
- Complete activity and responsibility history
- Notifications for unresolved or overdue actions

Convergence: Web + Cybersecurity + Cloud.

ChatGPT can make mistakes. Check important info.

4. Cloud Computing & Distributed Systems

1.ElasticAIInferenceOrchestrationUnderCostandLatency Constraints

BriefExplanation / Real-WorldContext

AI systems increasingly serve multiple models with different accuracy, latency and compute requirements. A naive deployment can overload expensive accelerators or violate response-time requirements during demand spikes. The challenge is dynamically deciding where and how inference should run.

Core Challenge

Build a distributed inference orchestrator that routes requests among heterogeneous workers according to latency, resource availability, model size, confidence and cost. Demonstrate behaviour under workload spikes and worker failure.

Expected 24-Hour Prototype

Expected prototype: model registry; request router; worker pool; queueing; autoscaling simulation; latency/cost telemetry; fallback strategy; inference result API; scheduling policy comparison.

Scope, Constraints & Convergence

Use containerized or simulated workers if GPUs are unavailable. The point is the orchestration algorithm and system behaviour, not provisioning a production Kubernetes cluster.

Convergence: AI supplies models; Cybersecurity supplies model/API authorization; Web provides

workloadand performancecontrols.

2.Geo-Distributed Disaster Data Replication with Graceful Degradation

BriefExplanation/Real-World Context

Critical systems cannot assume that a single region or network path will always be available.During a disaster, the system may need to continue accepting reports while disconnected from its primary service and later reconcile conflicting updates.

Core Challenge

Design a multi-region data service prototype with local write capability, synchronization after partition and a defined conflict-resolution policy. Demonstrate what happens when regions disagree and connectivity is restored.

Expected 24-Hour Prototype

Expected prototype: two or more service replicas; local persistence; replication protocol; partition simulation; conflict resolution; recovery; consistency-status API; operational dashboard.

Scope, Constraints & Convergence

The objective is not global-scale deployment. Teams should explicitly choose and defend their consistency model.

Convergence: AI can prioritize incoming reports; Cybersecurity protects replication and identity; Web provides incident operations.

3.Multi-Tenant Isolation and Noisy-Neighbor Defense Platform

Brief Explanation / Real-World Context
Shared cloud infrastructure creates a difficult fairness problem: one customer or workload can consume disproportionate CPU, memory, network or database capacity and degrade other tenants. Strong isolation must coexist with efficient resource utilization.

Core Challenge
Create a multi-tenant service that enforces quotas and adaptive resource policies while measuring tenant-level latency and resource consumption. Demonstrate the system under deliberately abusive workload patterns.

Expected 24-Hour Prototype
Expected prototype: tenant identity; resource quotas; rate limiting; workload scheduler; usage metrics; noisy-neighbor detector; adaptive throttling; admin API/dashboard.

Scope, Constraints & Convergence
Use local containers/processes or simulated resources. Do not equate a simple API rate limit with full tenant isolation.

Convergence: Cybersecurity supplies tenant identity/policy; AI can predict overload; Web provides operations and billing-like views.